

warning, and preserves lateral control effectiveness at high angles of attack. Also, the airplane must maintain positive static longitudinal stability at high lift coefficients and should demonstrate satisfactory stall recovery characteristics.

In order to visualize the principal effects of an airplane entering a spin, suppose the airplane is subjected to the rolling and yawing velocities shown in figure 4.32. The yawing velocity to the right tends to produce higher local velocities on the left wing than on the right wing. The rolling velocity tends to increase the angle of attack for the downgoing right wing (α_r) and decrease the angle of attack for the upgoing left wing (α_l). At airplane angles of attack below the stall this relationship produces roll due to yaw, damping in roll, etc., and some related motion of the airplane in unstalled flight. However, at angles of attack above the stall, important changes take place in the aerodynamic characteristics.

Figure 4.32 illustrates the aerodynamic characteristics typical of a conventional airplane configuration, i.e., moderate or high aspect ratio and little—if any—sweepback. If this airplane is provided a rolling displacement when at some angle of attack above the stall, the upgoing wing experiences a decrease in angle of attack with a corresponding increase in C_L and decrease in C_D . In other words, the upgoing wing becomes less stalled. Similarly, the downgoing wing experiences an increase in angle of attack with a corresponding decrease in C_L and increase in C_D . Essentially, the downgoing wing becomes more stalled. Thus, the rolling motion is aided rather than resisted and a yawing moment is produced in the direction of roll. At angles of attack below stall the rolling motion is resisted by damping in roll and adverse yaw is usually present. At angles of attack above the stall, the damping in roll is negative and a rolling motion produces a rolling moment in the direction of the roll. This negative

damping in roll is generally referred to as "autorotation."

When the conventional airplane is stalled and some rolling-yawing displacement takes place, the resulting autorotation rolling moments and yawing moments start the airplane into a self-sustaining rolling-yawing motion. The autorotation rolling and yawing tendencies of the airplane at high angles of attack are the principal prospin moments of the conventional airplane configuration and these tendencies accelerate the airplane into the spin until some limiting condition exists. *The stabilized spin is not necessarily a simple steady vertical spiral but may involve some coupled unsteady oscillatory motion.*

An important characteristic of the more conventional airplane configuration is that the spin shows a predominating contribution of the autorotation tendency. Generally, the conventional configuration has a spin motion which is primarily rolling with moderate yaw. High directional stability is favorable since it will limit or minimize the yaw displacement of the spinning airplane.

The fundamental requirement of the spin is that the airplane be placed at an excessive angle of attack to produce the autorotation rolling and yawing tendencies. Generally speaking, the conventional airplane must be stalled before a spin can take place. This relationship establishes a fundamental principle of recovery—the airplane must be unstalled by decreasing the wing angle of attack. The most effective procedure for the conventional configuration is to use opposite rudder to stop the sideslip, then lower the angle of attack with the elevators. With sufficient rudder power this procedure will produce a positive recovery with a minimum loss of altitude. Care should be taken during pullout from the ensuing dive to prevent excessive angle of attack and entry into another spin.

It should be appreciated that a spin is always a possible corollary of a stall and the self-sustaining motion of a spin will take place at